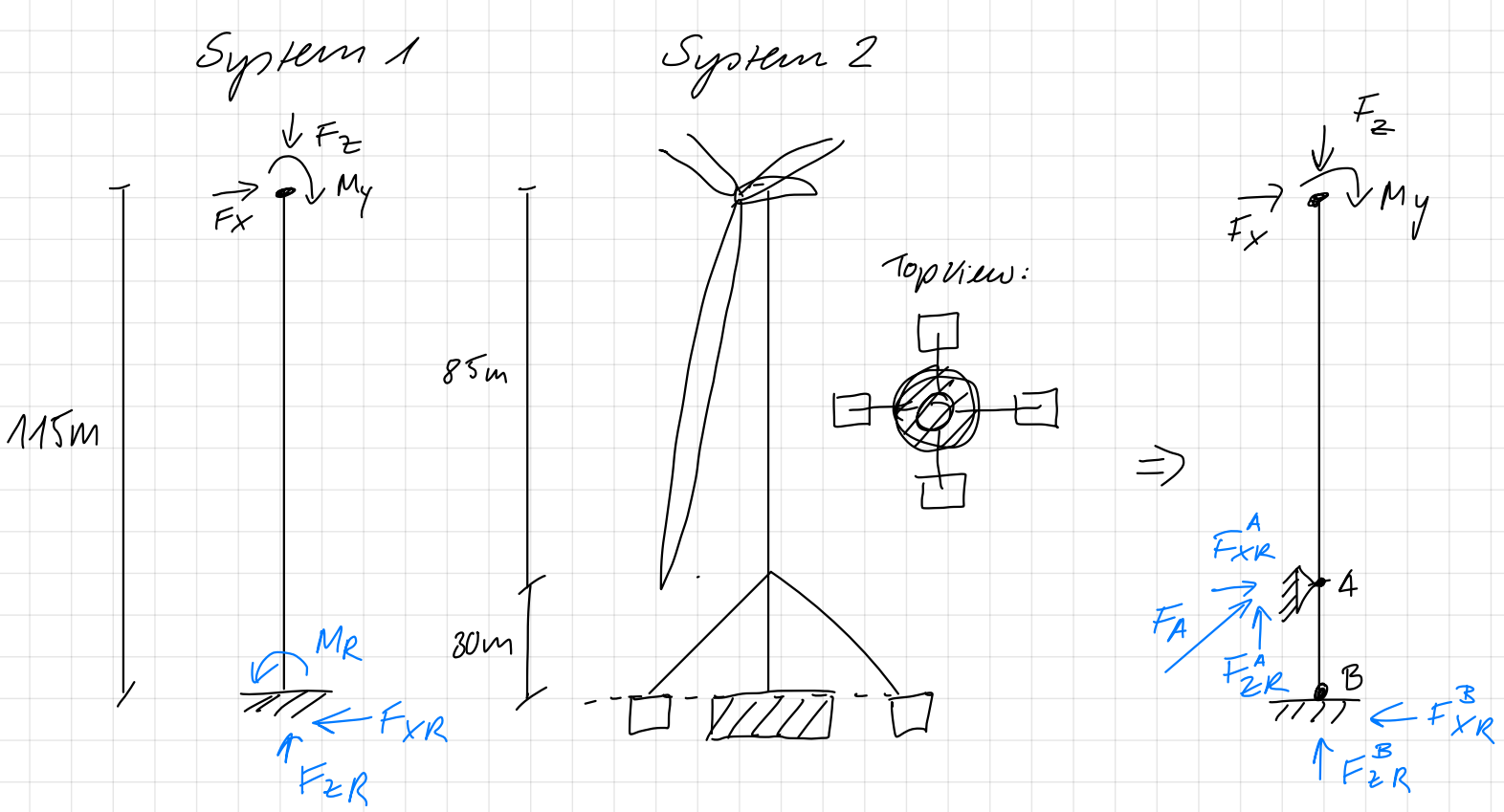


Calculation with $F_z = 5000\,kN$ $F_x = 2398\,kN$ $M_y = 38787\,kNm$

System 1:

$M_R = F_x \cdot 115m + M_y = 314057\,kNm$ $\quad F_{xR} = F_x \quad F_{zR} = F_z$

System 2:

$\Sigma M_B \curvearrowright^+ = F_x \cdot 115m + F_{xR}^A \cdot 30m + M_y = 0$

$\rightarrow \quad F_{xR}^A = -10468.6\,kN$

$\Sigma M_A \curvearrowright^+ = F_x \cdot 85m + 30 \cdot F_{xR}^B + M_y = 0$

$\rightarrow \quad F_{xR}^B = -8070.6\,kN$

Angle of $F_A = 45°$:

$F_{xR}^A = F_{zR}^A = -10468.6\,kN$

$+\uparrow \Sigma F_z = F_{zR}^A + F_{zR}^B - F_z = 0$

$\rightarrow F_{zR}^B = 15468.6\,kN$

$F_A = \sqrt{2} \cdot F_{xR}^A = 14804.8\,kN \rightarrow$ Search for ropes